Semi-supervised double robust test in kink regression model with treatment effect

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Abstract

Keywords:

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1 Introduction

2 Model

Let Y denote the response variable of interest, X be a univariate threshold variable, $A \in \{0, 1\}$ a binary treatment indicator, and W a p-dimensional vector of additional covariates that may affect the outcome.

We consider a semi-supervised setting where the observed data consist of a labeled dataset $\mathcal{L}$ and an unlabeled dataset $\mathcal{U}$. The labeled dataset contains n independent observations $\mathcal{L} = \{Z_i = (X_i, W_i^{\mathrm{T}}, A_i, Y_i)^{\mathrm{T}} : i = 1, \ldots, n\}$. The unlabeled dataset contains N independent observations $\mathcal{U} = \{(X_j, W_j^{\mathrm{T}}, A_j)^{\mathrm{T}} : j = n+1, \ldots, n+N\}$, for which the response variable Y is missing. We assume that data in both $\mathcal{L}$ and $\mathcal{U}$ are drawn from the same underlying distribution. In a typical semi-supervised scenario, the unlabeled sample size is much larger than the labeled sample size, i.e., $N \gg n$.

We consider a regression model that allows for a single unknown kink in the effect of the treatment across the running variable X. Specifically, the model is defined as

$$Y = \alpha_0 + \alpha_1 X + \tau A(X - \delta)\mathbf{1}(X \geq \delta) + \mu(W) + \varepsilon, \tag{1}$$

where α_0 is the intercept, α_1 is the baseline slope of X in the absence of treatment, τ represents the magnitude of the kink effect specific to the treated group, δ is the unknown threshold (kink point) at which the treatment effect may change, and $\mu(W)$ is an unknown smooth function capturing the effect of covariates W. The error term ε is assumed to satisfy the standard exogeneity condition $E[\varepsilon \mid A, X, W] = 0$.

The term $\tau A(X - \delta)\mathbf{1}(X \geq \delta)$ explicitly models the heterogeneous treatment effect as a function of the running variable X. It ensures that the kink effect only applies to the treated group ($A = 1$) and only becomes active when X exceeds the unknown threshold δ. In this formulation, the model can capture scenarios in which the treatment effect is negligible or absent below the threshold but increases (or decreases) linearly beyond it, reflecting a structural change in the treatment response.

Our primary interest lies in testing whether the treatment exhibits a heterogeneous effect characterized by a structural change (kink) at some unknown value of the running variable.

Formally, this corresponds to testing the null hypothesis that the treatment effect is homogeneous across all values of X, against the alternative that a kink exists at some unknown point:

$$H_0 : \tau = 0 \quad \text{versus} \quad H_a : \tau \neq 0. \tag{2}$$

Under the null hypothesis, the model reduces to a standard linear regression with a homogeneous treatment effect that does not vary with X, while under the alternative, there exists a subgroup of individuals for whom the treatment effect changes systematically once X crosses the threshold δ.

3 Methods

3.1 Semi-Supervised Doubly Robust Score Test

In a fully supervised setting where all responses are observed, testing the null hypothesis $H_0 : \tau = 0$ can be formulated using a doubly robust score function. Following standard semiparametric theory, the core component of the score evaluated at a candidate threshold δ for the i-th individual is given by

$$\psi_1(Z_i; \delta, \eta, \zeta) = \Big\{(X_i - \delta)\mathbf{1}(X_i \geq \delta)\Big\}\Big\{A_i - \pi(X_i, W_i; \eta)\Big\}\Big\{Y_i - h(X_i, W_i; \zeta)\Big\}, \tag{3}$$

where $\pi(X_i, W_i; \eta) = P(A_i = 1 \mid X_i, W_i)$ is the propensity score model, and $h(X_i, W_i; \zeta) = \mathbb{E}[Y_i | X_i, W_i]$ is the baseline outcome regression model under H_0. The construction in (3) guarantees double robustness: the estimating equation has expectation zero under H_0 if either the propensity score model or the baseline outcome model is correctly specified.

However, relying solely on the small labeled dataset $\mathcal{L}$ of size n yields a test statistic with substantial variance, leading to inadequate power for detecting structural changes. To fully exploit the massive unlabeled dataset $\mathcal{U}$ of size N, we propose an efficiency enhancement strategy that operates on two interconnected fronts: pooled estimation of the propensity score and nonparametric imputation of the unobserved scores.

First, we observe that the estimation of the propensity score model $\pi(X, W; \eta)$ depends only on the treatment assignment A and covariates (X, W), thus circumventing the need for the missing outcome Y. Consequently, we can estimate η by maximizing the binomial loglikelihood over the entire pooled dataset $\mathcal{L}\cup\mathcal{U}$ of size $M = n+N$. Let $\hat{\eta}_M$ denote this pooled estimator. Because $M \gg n$, the estimation error of $\hat{\eta}_M$ is asymptotically negligible compared to the usual $n^{-1/2}$ convergence rate, effectively eliminating a major source of variation in the score function. Conversely, the baseline outcome parameter ζ, which strictly requires Y, is estimated exclusively using the labeled data $\mathcal{L}$, yielding the estimator $\hat{\zeta}_n$.

Building upon the refined propensity score estimator, we next address the unobserved components of the score function for the unlabeled data. For the j-th unlabeled observation in $\mathcal{U}$, the response Y_j is missing. A statistically sound approach is to replace the unobservable Y_j in (3) with its conditional expectation $m(X, W; \zeta) = \mathbb{E}[Y|h(X, W; \zeta)]$. This substitution yields the conditional expected score for the unlabeled data:

$$\overline{\psi}_1(X_j, W_j, A_j; \delta) = \Big\{(X_j - \delta)\mathbf{1}(X_j \geq \delta)\Big\}\Big\{A_j - \pi(X_j, W_j; \hat{\eta})\Big\}\Big\{m(X_j, W_j; \zeta) - h(X_j, W_j; \zeta)\Big\}. \tag{4}$$

To nonparametrically estimate $m(X, W, A)$ while circumventing the curse of dimensionality, we employ a single-index Nadaraya-Watson (NW) kernel estimator. A natural and effective choice for the single index is the linear predictor derived from the baseline model, $S_i = h(X_i, W_i; \hat{\zeta}_n)$. Using the labeled dataset $\mathcal{L}$, the NW estimator evaluated at a given index s and treatment a is defined as

$$\widehat{m}(X, W; \zeta) = \frac{\sum_{i=1}^{n} Y_i K_{h_n}(h(X, W; \zeta) - h(X_i, W_i; \zeta))}{\sum_{i=1}^{n} K_{h_n}(h(X, W; \zeta) - h(X_i, W_i; \zeta))}, \tag{5}$$

where $K_{h_n}(\cdot) = h_n^{-1}K(\cdot/h_n)$ is a symmetric kernel density function with bandwidth h_n. Substituting (5) into (6) provides the imputed score $\widehat{\psi}_1(X_j, W_j, A_j; \delta)$ for each unlabeled observation, that is,

$$\widehat{\psi}_1(X_j, W_j, A_j; \delta) = \Big\{(X_j - \delta)\mathbf{1}(X_j \geq \delta)\Big\}\Big\{A_j - \hat{\pi}(X_j, W_j; \hat{\eta}_M)\Big\}\Big\{\hat{m}(X_j, W_j; \hat{\zeta}_M) - h(X_j, W_j; \hat{\zeta}_M)\Big\}, \tag{6}$$

where $\hat{\zeta}_M$ and $\hat{\eta}_M$ denote estimators of nuisance parameters obtained under the null model.

Specifically, these parameters satisfy the following estimating equations:

$$\Psi_2(\eta) = \frac{1}{n}\sum_{i=1}^{M} D_\eta(X_i, W_i)\Big\{A_i - \pi(X_i, W_i; \eta)\Big\} = 0,$$

$$\Psi_3(\zeta) = \frac{1}{n}\sum_{i=1}^{n} D_\zeta(X_i, W_i)\Big\{Y_i - h(X_i, W_i; \zeta)\Big\} + \frac{1}{N}\sum_{i=n+1}^{M} D_\zeta(X_i, W_i)\Big\{\widehat{m}(X_i, W_i; \zeta) - h(X_i, W_i; \zeta)\Big\} = 0,$$

where $D_\eta(X_i, W_i) = [\pi(X_i, W_i; \eta)\{1 - \pi(X_i, W_i; \eta)\}]^{-1}\partial\pi(X_i, W_i; \eta)/\partial\eta$ and $D_\zeta(X_i, W_i) = \partial h(X_i, W_i; \zeta)/\partial\zeta$ are the score-based weight matrices for the outcome and propensity score models, respectively. Notably, D_η incorporates the inverse variance weighting characteristic of the classical efficient score for logistic regression, ensuring efficient estimation of π under correct specification.

By strategically synthesizing the observed scores from $\mathcal{L}$ and the imputed scores from $\mathcal{U}$, we define the semi-supervised doubly robust score function as a weighted average:

$$\Psi_{SS}(\delta) = \frac{\lambda}{n}\sum_{i=1}^{n} \psi_1(Z_i; \delta, \hat{\eta}_M, \hat{\zeta}_n) + \frac{1-\lambda}{N}\sum_{j=n+1}^{n+N} \widehat{\psi}_1(X_j, W_j, A_j; \delta), \tag{7}$$

where $\lambda \in [0, 1]$ is a weight parameter controlling the relative contribution of the two datasets (for instance, setting $\lambda = n/M$ assigns equal weight to every observation).

To test H_0, we scan over a range of candidate threshold values and construct a supremum-type test statistic that maximizes the evidence against the null:

$$T_{n,SS} = \sup_{\delta\in\mathcal{D}} \left\{ \frac{\sqrt{n}}{\widehat{\sigma}_{SS}(\delta)} \Psi_{SS}(\delta) \right\}^2, \tag{8}$$

where $\widehat{\sigma}^2_{SS}(\delta)$ is a consistent estimator of the asymptotic variance of $\sqrt{n}\Psi_{SS}(\delta)$. Under the alternative hypothesis $H_a : \tau \neq 0$, the location exhibiting the greatest structural change can be consistently estimated by the maximizer of the standardized squared score, denoted by $\hat{\delta} = \arg\max_{\delta\in\mathcal{D}}\{n\Psi^2_{SS}(\delta)/\widehat{\sigma}^2_{SS}(\delta)\}$.

In practice, the search interval $\mathcal{D}$ is restricted to the interior of the support of X to ensure sufficient sample sizes on both sides of any candidate threshold. Taking advantage of the massive pooled dataset, we construct $\mathcal{D}$ as a trimmed set of the empirical quantiles of

$\{X_k\}_{k=1}^M$:

$$\mathcal{D} = \left\{\delta_{(k)} : k = \lceil M\theta \rceil, \ldots, \lfloor M(1-\theta) \rfloor \right\},$$

where $\delta_{(k)}$ is the k-th order statistic of the pooled threshold variable and θ is a small trimming constant (e.g., $\theta = 0.05$). This construction averts boundary effects and ensures robust nonparametric imputation.

3.2 Asymptotic Properties and Efficiency Gain

To establish the asymptotic properties of the semi-supervised test statistic $T_{n,SS}$, we must systematically account for the variation introduced by the nuisance parameters. Under the refined formulation, the estimation of both the propensity score parameter η and the baseline outcome parameter ζ leverages the massive unlabeled dataset. Let $\hat{\eta}_M$ and $\hat{\zeta}_M$ denote the solutions to the pooled and semi-supervised estimating equations $\Psi_2(\eta) = 0$ and $\Psi_3(\zeta) = 0$, respectively, and let $\hat{m}(\cdot)$ be the single-index NW estimator.

Because $\hat{\eta}_M$ is estimated using the pooled dataset, its convergence rate is $O_p(M^{-1/2})$, which becomes asymptotically negligible when scaled by $\sqrt{n}$ if $N \gg n$. The estimator $\hat{\zeta}_M$, while $\sqrt{n}$-consistent due to its reliance on the labeled data through $\hat{m}(\cdot)$, achieves higher efficiency than its fully supervised counterpart. We let $\psi^*_{SS}(Z_i; \delta)$ denote the *semi-supervised adjusted influence function*, which rigorously projects out the leading-order estimation errors of $(\hat{\eta}_M, \hat{\zeta}_M, \hat{m})$ from the score function.

To ensure uniform asymptotic validity, we impose the following regularity conditions:

(C1) The estimating equations $\Psi_2(\eta) = 0$ and $\Psi_3(\zeta) = 0$ admit unique solutions η_0 and ζ_0 in compact subsets of their respective parameter spaces.

(C2) The estimators $\hat{\eta}_M$ and $\hat{\zeta}_M$ are asymptotically linear, and their scaled asymptotic variances are finite and positive definite.

(C3) The conditional variance $\mathrm{Var}(Y \mid h(X, W; \zeta_0))$ is uniformly bounded. The single index $S = h(X, W; \zeta_0)$ has a continuous and bounded probability density function.

(C4) For any candidate threshold δ in the interior of $\mathcal{D}$, the probability $P(X \geq \delta)$ is strictly bounded away from 0 and 1.

(C5) The sample size ratio $n/N \to \rho \in [0, \infty)$ as $\min(n, N) \to \infty$.

(C6) The kernel function $K(\cdot)$ is a bounded, symmetric probability density. The bandwidth h_n satisfies $h_n \to 0$, $nh_n \to \infty$, and $nh_n^4 \to 0$ to prevent asymptotic bias from dominating the nonparametric estimation.

(C7) The class of functions $\{\psi^*_{SS}(\cdot; \delta) : \delta \in \mathcal{D}\}$ is P-Donsker.

Remark. Conditions (C1)–(C4) and (C7) are standard requirements for semiparametric doubly robust inference and empirical processes, adapted to the newly proposed semi-supervised estimating equations. Condition (C5) accommodates various missing data scenarios, seamlessly capturing the extreme case where the unlabeled data dominates ($\rho = 0$). Condition (C6) provides standard bandwidth requirements for the Nadaraya-Watson estimator acting on the dimension-reduced index $h(X, W; \zeta)$ to ensure uniform convergence.

Theorem 1 *Under assumptions (C1)–(C7), suppose that at least one of the nuisance models, $\pi(X, W; \eta)$ or $h(X, W; \zeta)$, is correctly specified. Then, under the null hypothesis $H_0 : \tau = 0$, the semi-supervised test statistic satisfies*

$$T_{n,SS} \xrightarrow{d} \sup_{\delta \in \mathcal{D}} G^2_{SS}(\delta), \quad \textit{as } \min(n, N) \to \infty,$$

where $\{G_{SS}(\delta) : \delta \in \mathcal{D}\}$ is a mean-zero Gaussian process with covariance function

$$\mathrm{Cov}(G_{SS}(\delta_1), G_{SS}(\delta_2)) = \frac{\mathbb{E}[\psi^*_{SS}(Z; \delta_1)\psi^*_{SS}(Z; \delta_2)]}{\sqrt{\mathbb{E}[\psi^{*2}_{SS}(Z; \delta_1)]\mathbb{E}[\psi^{*2}_{SS}(Z; \delta_2)]}}.$$

Theorem 1 confirms that the proposed test maintains the doubly robust property and proper Type I error control under the semi-supervised setting. The limiting distribution is the supremum of a Gaussian process, where its covariance structure incorporates the variability from the novel semi-supervised estimating equation $\Psi_3(\zeta)$ and the nonparametric imputation over $\mathcal{U}$.

The fundamental motivation for incorporating the unlabeled dataset is to improve testing power by drastically reducing the variance of the score function. Let $\Sigma_1(\delta) = \lim_{n\to\infty} \mathrm{Var}(\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi^*_1(Z_i; \delta))$ denote the asymptotic variance of the fully supervised score, and $\Sigma_{SS}(\delta) = \lim_{n\to\infty} \mathrm{Var}(\sqrt{n}\Psi_{SS}(\delta))$ denote that of the proposed semi-supervised score.

Theorem 2 *Under the same conditions as in Theorem 1, and utilizing the weighting parameter $\lambda = \rho/(1+\rho)$ (where $\lambda \to 0$ if $\rho = 0$), the asymptotic variance of the semi-supervised score function satisfies*

$$\Sigma_{SS}(\delta) = \Sigma_1(\delta) - \Delta(\delta),$$

where $\Delta(\delta) \geq 0$. Furthermore, $\Delta(\delta) > 0$ strictly holds provided that $\mathrm{Var}\big(\mathbb{E}[\psi_1(Z;\delta) \mid h(X,W;\zeta_0)]\big) > 0$.

Theorem 2 rigorously establishes the theoretical efficiency gain. By leveraging the semi-supervised estimator $\hat{\zeta}_M$ and nonparametrically imputing the score for the unlabeled data using the conditional expectation $\mathbb{E}[Y \mid h(X,W;\zeta)]$, we extract valuable structural information embedded in the massive covariate distribution. This mechanism projects the original score onto a subspace with inherently lower variability, yielding a strictly tighter variance $\Sigma_{SS}(\delta) < \Sigma_1(\delta)$.

To mathematically quantify how this variance reduction translates into enhanced test power, we investigate the test's behavior under local alternatives.

Theorem 3 *Under the conditions of Theorem 1, consider a sequence of contiguous local alternatives*

$$H_{1n} : \tau = n^{-1/2} c_0, \quad c_0 \neq 0.$$

Then, as $\min(n,N) \to \infty$, the test statistic $T_{n,SS}$ converges in distribution to

$$T_{n,SS} \xrightarrow{d} \sup_{\delta \in \mathcal{D}} \left[G_{SS}(\delta) + \mu_{SS}(\delta)\right]^2,$$

where $G_{SS}(\delta)$ is the Gaussian process defined in Theorem 1, and the non-centrality parameter (drift function) $\mu_{SS}(\delta)$ is given by

$$\mu_{SS}(\delta) = \frac{c_0\,\mathbb{E}\big[\pi(X,W)\{1-\pi(X,W)\}(X-\delta)(X-\delta_0)\mathbf{1}(X \geq \max\{\delta,\delta_0\})\big]}{\sigma_{SS}(\delta)},$$

with δ_0 denoting the true kink location and $\sigma_{SS}(\delta) = \sqrt{\Sigma_{SS}(\delta)}$.

Theorem 3 provides a profound insight into the mechanics of the proposed method. The numerator of the drift function $\mu_{SS}(\delta)$ remains mathematically identical to that of the fully supervised test, representing the true signal magnitude of the structural change. However, because Theorem 2 guarantees that $\sigma_{SS}(\delta) < \sigma_1(\delta)$, the non-centrality parameter $\mu_{SS}(\delta)$ is strictly inflated compared to its fully supervised counterpart. This deterministic upward shift in the drift function significantly increases the probability of the supremum statistic exceeding the critical value, theoretically ensuring a substantially higher asymptotic power.

3.3 Computation of the Critical Value

The asymptotic null distribution of the semi-supervised test statistic $T_{n,SS}$ established in Theorem 1 is the supremum of a complex Gaussian process. Because its covariance structure depends intricately on the unknown data-generating mechanism and the semiparametric imputation over $\mathcal{U}$, analytical derivation of the exact critical value is intractable. Furthermore, implementing a traditional nonparametric bootstrap would require repeatedly solving the pooled estimating equations $\Psi_2(\eta) = 0$ and $\Psi_3(\zeta) = 0$ for $\hat{\eta}_M$ and $\hat{\zeta}_M$, as well as recalculating the single-index NW kernel estimator $\hat{m}(\cdot)$ across every iteration. When the unlabeled sample size N is massive, this repeated evaluation renders standard bootstrap methods computationally prohibitive.

To overcome this computational bottleneck, we propose a highly efficient multiplier bootstrap (perturbation resampling) method to approximate the limiting null distribution. Let $\widehat{\psi}^*_{SS}(O_k; \delta)$ denote the empirical estimate of the adjusted influence function for the k-th observation ($k = 1, \ldots, M$), where O_k represents the available data for either labeled subjects ($k \leq n$) or unlabeled subjects ($k > n$). This empirical adjusted influence function mathematically projects out the first-order estimation errors of the semi-supervised nuisance parameters $(\hat{\eta}_M, \hat{\zeta}_M)$ and the nonparametric imputation $\hat{m}(\cdot)$ (explicit projection formulas and matrix definitions are deferred to the Appendix). The variance estimator $\widehat{\sigma}^2_{SS}(\delta)$ is consistently computed using the empirical sample variance of these adjusted influence functions.

We generate a sequence of independent standard normal random variables, $\xi = (\xi_1, \ldots, \xi_M)^\top$, which are strictly independent of the observed data. The perturbed semi-supervised test statistic is constructed as:

$$T^*_{n,SS} = \sup_{\delta \in \mathcal{D}} \frac{n}{\widehat{\sigma}^2_{SS}(\delta)} \left\{ \frac{\lambda}{n} \sum_{i=1}^{n} \xi_i \widehat{\psi}^*_{SS}(Z_i; \delta) + \frac{1-\lambda}{N} \sum_{j=n+1}^{M} \xi_j \widehat{\psi}^*_{SS}(X_j, W_j, A_j; \delta) \right\}^2. \tag{9}$$

By repeatedly generating independent sets of $\{\xi_k\}_{k=1}^{M}$, we can simulate a large number of perturbed test statistics. The empirical distribution of $T^*_{n,SS}$ provides a computationally efficient and statistically consistent approximation of the limiting distribution of $T_{n,SS}$ under the null hypothesis. The validity of this multiplier resampling algorithm is formally guaranteed by the following theorem.

Theorem 4 *Under the null hypothesis and conditions (C1)–(C7), conditionally on the observed pooled dataset $\mathcal{L} \cup \mathcal{U}$, we have*

$$\sup_{x\in\mathbb{R}} \left| P^*(T^*_{n,SS} \leq x) - P(T_{n,SS} \leq x) \right| \xrightarrow{p} 0,$$

where P^* denotes the probability measure induced by the multiplier bootstrap perturbations ξ, and $T^*_{n,SS}$ is the perturbed test statistic defined in (9).

Theorem 4 ensures that the upper α-quantile of $T^*_{n,SS}$, denoted as C_α, serves as an asymptotically exact critical value. An α-level test safely rejects the null hypothesis H_0 if the observed statistic $T_{n,SS} > C_\alpha$.

To facilitate the practical implementation of our approach, the search space $\mathcal{D}$ is approximated by an equally spaced grid derived from the empirical quantiles of the pooled threshold variable $\{X_k\}_{k=1}^{M}$. The entire procedure for computing the p-value is summarized in Algorithm 1.

4 Simulation Studies

5 A Case Study

6 Conclusion Remarks

Appendix : Proofs of Theorems

Algorithm 1 Multiplier Bootstrap for the Semi-Supervised Doubly Robust Test

Require: Labeled data $\mathcal{L}$, unlabeled data $\mathcal{U}$, pooled sample size $M = n+N$, threshold search space $\mathcal{D}$, fitted semi-supervised nuisance estimators $(\hat{\eta}_M, \hat{\zeta}_M, \hat{m})$, and a large integer B.

Ensure: The estimated p-value.

1: Calculate the observed test statistic $T_{n,SS}$ using (8).

2: Evaluate the empirical adjusted influence functions $\widehat{\psi}^*_{SS}(O_k; \delta)$ for all $k = 1, \ldots, M$ over the grid $\mathcal{D}$.

3: **for** $b = 1, 2, \ldots, B$ **do**

4: *Step 1.* Draw a perturbation vector $\xi^{(b)} = (\xi_1^{(b)}, \ldots, \xi_M^{(b)})^\top$ from the standard multivariate normal distribution $\mathcal{N}(0, I_M)$.

5: *Step 2.* Calculate the perturbed semi-supervised test statistic:

$$T^{*(b)}_{n,SS} = \sup_{\delta \in \mathcal{D}} \frac{n}{\widehat{\sigma}^2_{SS}(\delta)} \left\{ \frac{\lambda}{n} \sum_{i=1}^{n} \xi_i^{(b)} \widehat{\psi}^*_{SS}(Z_i; \delta) + \frac{1-\lambda}{N} \sum_{j=n+1}^{M} \xi_j^{(b)} \widehat{\psi}^*_{SS}(X_j, W_j, A_j; \delta) \right\}^2 .$$

6: **end for**

7: *Step 3.* Calculate the p-value based on the empirical distribution of the bootstrapped statistics:

$$p\text{-value} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\left(T^{*(b)}_{n,SS} > T_{n,SS} \right).$$

References